Gustavo Landfried
Chemin des Bégonias 1
1018 Lausanne

April, 2026

Dear Sir or Madam,

I am a Swiss citizen based in Lausanne, holding a PhD in Computer Science with a specialisation in Bayesian causal inference and a degree in Anthropological Sciences. I relocated to the Suisse romande in April 2026 after several years in Argentina, where I developed my research, teaching and consulting activities. I currently work as a Senior Data Scientist in remote mode for Latin American multinationals. This combination of disciplines is not accidental: it is a trajectory built deliberately to operate at the intersection of statistical rigour and the understanding of social, economic and technological phenomena.

In recent years I have led statistical projects with multiple institutional stakeholders. For the *Foundation for Innovative New Diagnostics* (FIND) and Fiocruz, I conducted the statistical evaluation of Chagas diagnostic tests in four countries using B-LCA models; in doing so, I identified a methodological error in the European consortium COST Action CA18208, which led to a publication in *PLoS Neglected Tropical Diseases* (2024). I have developed hierarchical Bayesian models for commercial policy decisions at Coca-Cola FEMSA and KAVAK. My publications in the *Journal of Statistical Software* (2025), *Journal of Computational Social Science* (2025) and *PLoS ONE* (2019) reflect the ability to combine methodological rigour with communicable results. I am proficient in R (Stan), Python (PyMC, Pyro, PyTorch), Julia (Turing) and large-scale data architectures (Delta Lakehouse, Spark).

As a university lecturer in Bayesian Causal Inference (UNSAM/UBA, 2024–present), I have developed an original eight-unit programme covering causal graphical model specification through to ergodic decision-making. As founder and coordinator of Bayes Plurinacional, I organise annual scientific events across Latin America (Santiago 2023, Salta 2024, Bogotá 2025, Ciudad del Este 2026). I have supervised 7 undergraduate theses in Computer Science and Data Science at the University of Buenos Aires.

Bayesian causal inference is the approach needed to quantify uncertainty and separate correlations from genuine causal effects. It has direct applications across multiple domains: public health, institutional statistics and foresight, data-scarce business decisions, academic research and specialised training. I am looking in the Suisse romande because this is the region where I am settling with my wife (we married in Lausanne 15 years ago) and where we intend to build our lives durably. I am available immediately, with no need to relocate.

Sincerely,

LANDFRIED

Gustavo Landfried

Causal Data Science

Landfried, Gustavo

Nationality	Argentine - Swiss
Languages	English C1 | Spanish native | French C1 | German A1
Residence	Lausanne (Swiss) | Buenos Aires (Argentina)
Birthday	October 1985
Telephone	+41 77 407 1874 | +54 911 5103 7334
Email	gustavolandfried@gmail.com
Web	glandfried.github.io/
Linkedin	linkedin.com/in/gustavo-landfried
Github	github.com/glandfried

PhD Computer Science. MSc Social Anthropology. **Bayesian Causal Inference**.

- Lead of statistical projects in international organizations and multinational corporations
- Researcher with publications in causal Bayesian inference (JSS, JCSS, PLoS ONE)
- Founder and coordinator of the Bayesian statistics community of Latin America
- Developer of custom statistical software tailored to each problem (Pyro, PyMC)
- Specialist in efficient statistical methods for detecting hidden variables
- Interdisciplinary flexibility and accessible communication for diverse audiences

Statistical professional in international organizations and corporations.

2025 – actual

Causal Data Scientist @ *Latin America*

○ **Coca-Cola (FEMSA)**. Analysis of heterogeneous causal effects in the Juntos+ loyalty promotion system. Problem: promotions were designed based on fixed rules. Result: development of the strategy adopted by the team, based on a causal Bayesian network.
○ **Automotive marketplace (KAVAK)**. Impact evaluation of marketing efforts. Problem: A/B Testing with low statistical significance and fixed-rule attribution. Result: Development of a data-efficient methodology for the analysis of causal experiments and Data-Driven attribution.
○ **Paid media optimizer (Muttdata)**. Budget allocation across advertising channels. Problem: Poor predictive performance for high-quality conversions (low frequency). Result: Development of a hierarchical Bayesian model that significantly improved impact predictions.

2023 – 2024

Responsible for the evaluation of diagnostic tests @ *Latin America and Europe*

○ **Foundation for Innovative New Diagnostics (FIND)**. Evaluation of rapid diagnostic tests for Chagas in South America. Problem: the current methodology, expensive and slow, is a bottleneck. Result: we showed that rapid tests allow the development of a more efficient and safe methodology.
○ **European Cooperation in Science and Technology (Action CA18208)**. "Novel tools for test evaluation and disease prevalence estimation". Problem: I reported to the European consortium an error they were making in the use of the Dendukuri model. Result: publication in *PLoS Neglected Tropical Diseases*.
○ **National Health Administration (Malbrán)**. Impact experiment of Chagas strains on diagnostics. Problem: low statistical significance due to sparse data. Result: high significance achieved through a causal Bayesian model; publication in the *Journal of Molecular Diagnostics*.

2022 – actual

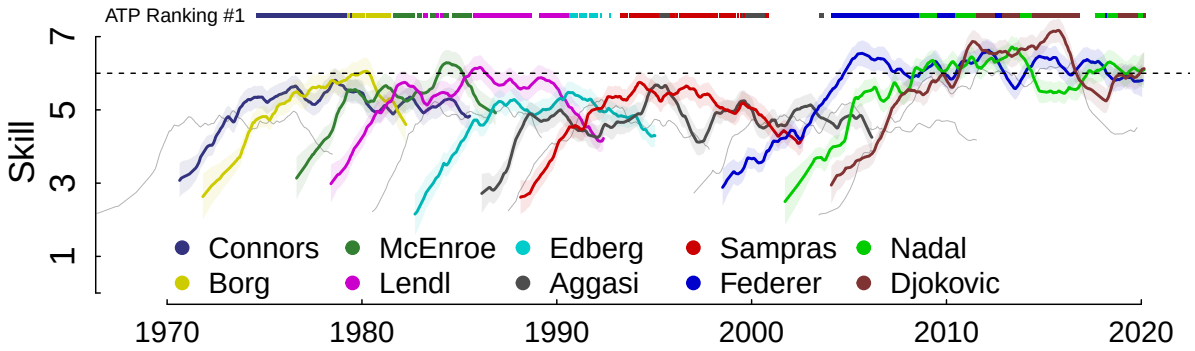
Promoter of the continental Bayesian statistics community @ *Latin America*

○ **Plurinational Bayes**. Problem: the adoption of Bayesian inference is lagging at the continental level. Result: continuous organization of Artificial Intelligence events and human resources training in Bayesian statistique, Santiago 2023, Salta 2024, Bogotá 2025, Ciudad del Este 2026.
○ **Professional training in Bayesian causal inference**. Separating correlation from causality and finding optimal decisions over time. Full content in "teaching" section.

Optimal reasoning system in contexts of uncertainty

2021 – 2022 The state-of-the-art skill estimator @ *Journal of Statistical Software*

Efficiently inference in causal networks with millions of nodes and irregular structures through distributed message-passing algorithms and analytical approximation methods.



2019 – 2020 Impact of the social network on skills @ *Journal of Computational Social Sicence*

Impact study of social structure on individuals' skill. Problem: classical socio-cultural studies could not evaluate the impact of social structure on human knowledge. Result: massive data from the OGS (Online Go Platform) community, graph analysis (popularity, closeness, betweenness), and the state-of-the-art skill estimator allowed for the evaluation of this impact.

2017 – 2018 Impact of team formation on skill @ *PLoS ONE*

We analyzed a community of the game Risk, between 2006 and 2013, where players could play in teams. We showed that team play leads to higher skill levels in the long term, and that playing with a stable partner accelerates skill acquisition.

Professional in social anthropology with a quantitative orientation

2015 – 2016 Social Analysis Program of Latin American Citizenship @ *PASCAL*

Main entity created by the Media Law in Argentina for television audience measurement (ratings). My role within PASCAL was twofold: coordinator between the technical team and the social team and administrator of the database and the automated telephone survey system.

2014 Training for admission to the PhD in Computer Science @ *Universidad de Buenos Aires*

2012 – 2013 Union of Popular Economy Workers @ "Argentina Trabaja" Program

Assistant of the former president of the National Institute of Industrial Technology 2002-2011. My work was focused on the worker cooperatives of "Argentina Trabaja". My tasks were directed by Enrique Palmeiro, assistant of Francisco, who was appointed Pope while working with him. We conducted evaluations of the cooperatives to promote their production.

Society and culture viewed as complex systems

2011 – 2012 Prediction of parliamentary alliances @ *Université de Lausanne*

Course at the Institute of Applied Mathematics of the Faculty of Social and Political Sciences at the University of Lausanne on complexity and agent-based models, taught by Jorge Peña. Master's thesis implementing Robert Axelrod's alliance prediction model applied to European parliamentary democracies.

2010 Education in exact and natural sciences @ *Universidad de Buenos Aires*

2008 – 2009 Anthropology, complexity, simulation, and ethnography @ *Grupo Antropocaos*

Research assistant in the UBACyT project "Simulation models and anthropology, their relationship with ethnographic work". Teaching assistant in the seminar "Artificial societies and ethnography".

Education Buenos Aires University

2016 – 2022 PhD in **Computer Science.**

2012 – 2014 Licentiate in Computer Science. (Suspended after promotion to PhD)

2005 – 2009 Licentiate (BSc + MSc) in **Anthropological Sciences**.

Teaching

2024 – actual *Bayesian Causal Inference @ Laboratorios de Métodos Bayesianos - ECyT UNSAM*

(1) Specification and evaluation of alternative causal models. (2) Exact and approximate inference (message-passing, variational, MCMC, SBI). (3) Inference flow, backdoor criterion, and Do-Calculus. (4) Causal ML algorithms and frameworks. (5) Temporal algorithms and generative temporal models. (6) Counterfactual inference and direct causal effects. (7) Decision-making, optimization as inference, and ergodic utility theory. (8) Socioeconomic interventions and productive financial instruments.

2019 – actual *One-on-one mentoring.* Director of 7 master's thesis in Computer Science and Data Science.

2016 – 2024 Previous courses at UBA (Depts of Computer Science and Anthropology): *Bayesian Inference*, *Algorithms & Data Structures* (I & II), *Introduction to CS*, *Functional Programming* (`Haskell`), *Computational Social Science*.

Scientific research

Articles

○ **Landfried** G., Mocskos E. *TrueSkill Through Time: reliable initial skill estimates and historical comparability in Julia, Python and R*. **Journal of Statistical Software. 2025**. 10.18637/jss.v112.i06

○ **Landfried**, G., Cairo G., Mocskos E. *Network Position and Learning Dynamics: Unveiling the Impact of Social Structure on Skill Acquisition in Online Gaming Platforms*. **Journal of Computational Social Sicence. 2025** 10.1007/s42001-025-00370-2.

○ Longhi, SA; Muñoz-Calderón, A; García-Casares, L; Irazu, L; Rodríguez, MA; **Landfried**, G; Alonso-Padilla, J; Schijman, AG; and Chagas-group *Inter-Laboratory Harmonization Study and Prospective Evaluation of the PURE-T. cruzi-LAMP Assay for Detecting Parasite Presence in Newborn Dried Blood Spots*. **The Journal of Molecular Diagnostics. 2024.** 10.1016/j.jmoldx.2024.08.007

○ Denwood, M; Nielsen, S; Olsen, A; Jones, H; Coffeng, L; **Landfried**, G; Nielsen, M; Levecke, B; Thamsborg, SM; Eusebi, P; Meletis, E; Kostoulas, P; Hartnack, S; Erkosar, B; Toft, N. *All that glitters is not gold: an interpretive framework for diagnostic test evaluation using Ascaris lumbricoides as a conceptual example*. **Plos Neglected Tropical Disease. 2024**.10.1371/journal.pntd.0012481

○ **Landfried**, G; Fernandez Slezak, D; Mocskos, E: *Faithfulness-boost effect: Loyal teammate selection correlates with skill acquisition improvement in online games.* **PLoS one. 2019**. 10.1371/journal.pone.0211014

Skills

If there is balance, it weighs less.

Software tools

`Delta Lakehouse` (`DataBricks`, `Spark`, `Streaming`, `Pipeline` ...), `Agentic AI` (`Claude Code`, `MCP`, `Hooks`, `Subagents`, `Skills`, ...), `Python` (`Pytorch`, `TensorFlow`, `Keras`, `PyMC`, `Pyro`, `Particles`, `Scipy`, `Sklearn`, `Pandas`, `Numpy`, ...), `R` (`Stan`, `Tidyr`, ...), `Julia` (`Turing`, ...), `C++` (`MPI`), `C#` (`Infer.NET`, ...), `Java`, `Haskell`, `Bash` (`screen`, `ssh`, `vi`, `rsync`, `awk`, ...), `SQL`, `NoSQL`, `Git`, `Docker`, `Latex` (`Tikz`), `Html`, ...

REPÚBLICA ARGENTINA

Universidad de Buenos Aires

Facultad de Ciencias Exactas y Naturales

El Rector de la Universidad y el Decano de la Facultad

Por cuanto, el Licenciado en Ciencias Antropológicas (orientación Sociocultural) **Gustavo Andres Landfried** natural de la Ciudad de Buenos Aires, ha finalizado el 9 de abril de 2024 los estudios correspondientes al **Doctorado de la Universidad de Buenos Aires, área Ciencias de la Computación.**

Por lo tanto, de acuerdo con las normas vigentes en esta Universidad, le confieren el presente diploma de **Doctor de la Universidad de Buenos Aires, área Ciencias de la Computación.**

Buenos Aires, 20 de noviembre de 2024

Documento Nacional de Identidad 31916400

Título de la tesis: "Análisis bayesiano del aprendizaje en comunidades de video juegos."

Calificación Sobresaliente aprobada el 09/04/2024

LEONARDO FEDERICO GOTTIFREDI
DIRECTOR GENERAL DE TÍTULOS Y PLANES

MARÍA CATALINA NOSIGLIA
SECRETARIA DE ASUNTOS ACADÉMICOS

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Landfried, Gustavo Andres
DNI - 31916400
UBA - 294/21

Diploma

Ministerio de Capital Humano
Secretaría de Educación
Subsecretaría de Políticas Universitarias
Argentina

000001693103

REPÚBLICA ARGENTINA

Universidad de Buenos Aires

Facultad de Filosofía y Letras

El Rector de la Universidad y la Decana de la Facultad

Por cuanto, **Gustavo Andrés Landfried**

natural de la Ciudad de Buenos Aires, ha finalizado el 20 de mayo de 2013 los estudios correspondientes a la carrera de **Ciencias Antropológicas.**

Por lo tanto, de acuerdo con las normas vigentes en esta Universidad, le confieren el presente diploma de **Licenciado en Ciencias Antropológicas (orientación Sociocultural).**

Buenos Aires, 12 de agosto de 2014

SOFÍA IRENE THISTED
Secretaria de Asuntos Académicos

GRACIELA ALEJANDRA MORGADE
DECANA

ALBERTO E. BARBIERI
RECTOR

MARÍA CATALINA NOSIGLIA
SECRETARIA DE ASUNTOS ACADÉMICOS

Documento Nacional de Identidad 31916400

LEONARDO FEDERICO GOTTIFREDI
DIRECTOR GENERAL DE TÍTULOS Y PLANES

MINISTERIO DE EDUCACIÓN

DIRECCIÓN NACIONAL DE GESTIÓN UNIVERSITARIA

Certifico que las firmas del Rector, Dr. Alberto Edgardo BARBIERI, Secretaria de Asuntos Académicos, Lic. María Catalina NOSIGLIA y el Director General de Títulos y Planes, Sr. Leonardo Federico GOTTIFREDI que anteceden, guardan similitud con la/s que obran en nuestros registros.

Buenos Aires, - 3 NOV 2014

Cont. MARIA I. ILLAPRESCAPE
DIRECCIÓN NACIONAL DE GESTIÓN UNIVERSITARIA

LANDSRIEDA, GUSTAVO (21)

REPUBLICA ARGENTINA
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Firma del interesado

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Dirección Nacional de Gestión Universitaria
044081277
Landriaed, Gustavo Andrés
DNI 31916400
UBA - Licenciado en Ciencias Arts

Bayesian Causal Inference

IBC1.2026.1

Professor: Gustavo Landfried[1,2,3]

1. School of Science and Technology, Universidad Nacional de San Martín (UNSAM)
2. Bayesian Methods Laboratory
3. Muttdata

Contact: `glandfried@dc.uba.ar` `glandfried.github.io`

April 13, 2026

All data-driven sciences develop causal arguments to explain and predict the world. The evaluation of causal hypotheses is attracting increasing interest from industry, which needs to measure the real impact of its actions. The advantage of causal models lies in their predictive capacity, which adapts naturally to contextual changes such as human interventions. Moreover, we know that if causal arguments correspond to the underlying causal reality, no artificial intelligence model, however complex, can outperform them.

In this course we will review the foundations for evaluating alternative causal models M given data D, $P(M|D)$, both with and without interventions. We will also review methods for generating predictions of hypotheses H internal to models M, $P(H|D, M)$, both for evaluating out-of-sample causal effects and for predicting the impact of alternative counterfactual actions. Finally, we will address the problem of optimal decision-making as an inference problem in which we verify that none of the alternative counterfactual decisions improve the result on the target variable.

Faced with the dizzying pace of AI tools, this course prioritizes immutable foundations. Despite all advances, since the 18th century no new system for reasoning under uncertainty has been proposed. If machines ever surpass humans, they will have to do science by strictly applying probability rules to evaluate causal theories. To reach (intersubjective) truths under uncertainty, one simply needs to know how to apply and preserve the universal principle of not lying: do not assert more than is known, without hiding what is known. This yields optimal belief distributions given the available information, unimprovable in practical terms.

Objectives

This course focuses on evaluating alternative causal arguments through the (approximation to) strict application of probability rules (Bayesian approach), the system of reasoning under uncertainty. The main objective is to review methods developed in recent decades for:

- **Causal models.** Mathematically specifying causal arguments expressed in natural language through intuitive graphical methods.
- **Inference flow.** Understanding when inference flow opens and closes in causal structures, in order to separate spurious influence from causal influence.
- **Causal effect estimation.** Applying Machine Learning methods to identify the causal effect between pairs of variables from observational data without interventions.
- **Counterfactual predictions.** Specifying joint probability distributions between factual and counterfactual outcomes, and generative reasoning about counterfactual consequences.
- **Decision-making over time.** Selecting optimal decisions in action-perception cycles with a hidden (simulated) nature.

About the instructor

Gustavo Landfried is a professor of Bayesian Causal Inference at the School of Science and Technology of the Universidad Nacional de San Martín (UNSAM), and one of the leading figures in Bayesian causal inference education in Latin America. After obtaining a degree in Social Anthropology, Gustavo earned his PhD in Computer Science from the Universidad de Buenos Aires (UBA) under the supervision of the directors of the Applied Artificial Intelligence Laboratory (Diego Slezak) and the Interdisciplinary High-Performance Computing Laboratory (Esteban Mocskos). He specialized in Bayesian skill analysis for the video game industry, which led him to develop and maintain the state-of-the-art packages in Python, Julia and R (*TrueSkill Through Time*, Journal of Statistical Software, 2025). Gustavo currently works in industry (Muttdata) as a Data Scientist developing Bayesian and causal inference projects for large multinational corporations in Latin America. He is also co-founder and coordinator of the community `bayesplurinacional.org`, organizing annual scientific events across the continent (Santiago 2023, Salta 2024, Bogotá 2025, Ciudad del Este 2026). He has supervised seven BSc/MSc theses in Computer Science and Data Science.

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All sub-units include assessed assignments. Upon completion of each full unit, an evaluation is conducted based on multiple-choice questions with belief distributions.

1 Specification and Evaluation of Causal Arguments

System of reasoning under uncertainty, graphical specification of causal arguments, and evaluation of alternative causal models with and without interventions.

1.1 Alternative Causal Arguments and Uncertainty

Truth and falsehood. Intersubjective agreements: maximum uncertainty given available information. Honest belief distribution across parallel universes of causal arguments. Preservation of prior belief compatible with the data and prediction as a sum of parallel universes compatible with what one wants to predict. System of reasoning under uncertainty: the rules of probability. Graphical methods for mathematically specifying probabilistic causal arguments: Bayesian networks. Basic properties of graphical models: structure, independence and factorization. Measuring uncertainty over hypothesis spaces internal to models, $P(\text{Hypothesis}|\text{Data}, \text{Model})$. Evaluating alternative causal arguments from observational data without interventions, $P(\text{Model}|\text{Data})$. The evidence, $P(\text{Data}|\text{Model})$, and its transformations (geometric mean and logarithm).

Assignment 1.1: Specification and evaluation of Monty Hall and Base models.

1.2 Surprise: The Problem of Communication with Reality

The fundamental elements of the reasoning system under uncertainty (Kolmogorov and Rényi axioms). The invariant structure of empirical data. Its isomorphism with communication systems with reality. Surprise as a source of information. The equivalence between the geometric

mean of the prediction, $P(\text{Data}|\text{Model})$, and cross-entropy. The advantage of causal models that correspond to the underlying causal reality. The mathematical definition of the principle of not lying: maximum entropy given available information. Specification of causal models for contexts with corrupted data. Context-adaptive causal mechanisms: do-operator and gates. Specification of causal arguments through factor graphs. Identification of causal models with and without interventions. Evaluation of alternative measurement procedures.

Assignment 1.2: Specification and evaluation of the Monty Hall model with corrupted data.

2 Inference Methods and Probabilistic Programming

Exact inference methods through message passing between nodes of the causal network, and approximate inference — both analytic and sampling-based — through probabilistic programming languages.

2.1 Exact Inference and Message Passing

Exact inference. The variable elimination algorithm. Decomposition of probability rules as messages exchanged between nodes of causal networks: sum-product algorithm. The extended Monty Hall model case. Conjugate distributions. Beta-Bernoulli model and its applications in medicine and marketing. Univariate Gaussian models and the exact solution to the skill estimation model. Multivariate Gaussian models: posterior and evidence. Bayesian linear regression. Arbitrary hypothesis selection via maximum likelihood. Interpretation of estimated parameters under known causal realities. Evaluation of linear models of increasing complexity. Absence of overfitting through strict application of probability rules. Prediction based on model ensembles, $P(\text{Data})$.

Assignment 2.1: Evaluation of linear models of increasing complexity.

2.2 Forgetting: The Problem of Approximate Inference

Analytic approximation methods. KL divergence and approximate inference via expectation propagation. The case of skill estimation in the video game industry. Reversed KL divergence and approximate inference via variational inference. The Bayesian version of the recommendation system that won the Netflix competition. The probabilistic programming language `Pyro`. Sampling-based approximation methods. Monte Carlo approximate inference: Metropolis-Hastings, Hamiltonian Monte Carlo and Sequential Monte Carlo. Probabilistic programming languages `PyMC`, `Stan`, `Turing`. The case of evaluating alternative measurement instruments and the Bayesian p-value. Generative approximation methods. Approximate Bayesian Computation (ABC): likelihood-free inference via summary statistics and distance metrics. Current methods for estimating the joint distribution via neural networks from simulated data (SBI) and efficient inference.

Assignment 2.2: Evaluation of alternative measurement instruments: Chagas diagnostic test with corrupted data.

3 Causal Predictions

Elimination of spurious influence from predictions and the steps of causal inference: 1) specification of the causal structure, 2) identification of the estimand, 3) estimation of the causal effect.

3.1 Inference Flow and Elimination of Spurious Influence

Inference flow in causal networks. Opening and closing of inference flow in basic structures: pipe, fork and collider. Opening and closing of inference flow along paths of any length: d-separation. Elimination of spurious influence from predictions: backdoor controls. Good and bad controls. Neutral controls and heterogeneous causal effects. Use of linear regression for estimating non-linear causal effects. Interpretation of estimated coefficients. The Table 2 Fallacy. The steps of causal inference: 1) specification of the causal structure, 2) identification of the estimand, 3) estimation of the causal effect. Discovery of causal structures compatible with estimated joint probability distributions from data.

Assignment 3.1: Causal predictions under diverse underlying causal realities.

3.2 Estimands and Do-Calculus

The estimand for the general causal effect: adjustment formula and propensity score. The fundamental characteristic of randomized experiments: independence between treatment and its potential counterfactual outcomes. The "conditional ignorability" analogy as a criterion for selecting control variables in the potential outcomes approach to causal inference. Estimand of causal effects under conditional ignorability and deterministic consistency. The 3 rules of Do-Calculus: Rule 1, Ignoring an observation; Rule 2, Intervention as observation; Rule 3, Ignoring an intervention. Alternative estimand when hidden controls exist: frontdoor formula. Alternative estimand when randomized experiments fail: instrumental variables.

Assignment 3.2: Estimation of causal effects: general, frontdoor and instrumental.

4 The Algorithm Zoo

From early strategies to state-of-the-art supervised learning models with proven capacity for estimating heterogeneous causal effects in high-dimensional settings.

4.1 Traditional Methods

Traditional effect estimation methods. Comparison of treated and control units with similar characteristics: Matching. Use of the Propensity Score as a similarity measure. Reweighting the sample through the Propensity Score: Inverse Probability Weighting (IPW). Limitations of traditional methods in high-dimensional settings. The Causal Machine Learning (CausalML) solution: high-capacity predictive algorithms for estimating heterogeneous causal effects (CATE). The Python causal inference ecosystem: integration of `DoWhy`, `EconML` and `CausalML`. The concept of Meta-Learners: strategies for decomposing estimation of the conditional causal effect (CATE) into standard supervised prediction problems. S-Learner and T-Learner: advantages and disadvantages. The impact of lack of "positivity" on causal predictions: the personal lending industry case.

Assignment 4.1: Causal predictions under different degrees of positivity.

4.2 Leading Algorithms in Causal Inference Competitions

Fundamentals and application of the Doubly Robust Learner. Causal inference competitions at the Atlantic Causal Inference Conference. The regularization problem in causal effect estimation. Estimation strategy under partially linear models: orthogonalization. Fundamentals and application of Doubly Machine Learning: regularization in nuisance models and isolation of the causal effect. Non-linear algorithms based on tree ensembles. From Random Forests to Causal Forests. From Bayesian Additive Random Trees to Bayesian Causal Forests. The problem of estimated confidence intervals in data competitions and possible solutions. Hybrid algorithms.

Assignment 4.2: Causal inference competition ACIC 2022.

5 Time

Specialized algorithms for estimating causal effects in time series. Generative temporal causal models: specification, inference and evaluation.

5.1 Temporal Algorithms

The advantage of longitudinal data: time-invariant fixed effects. Classical Difference-in-Differences (DiD). The problem of DiD with heterogeneous effects. DiD with multiple periods and staggered adoption. New robust estimators: Callaway-Sant'Anna, Sun-Abraham, Doubly Robust DiD. The problem of absence of comparable control units. Construction of synthetic control groups through weighted combination of untreated units. Combining synthetic weights with DiD structure allowing staggered timing and heterogeneous trends: Synthetic Difference-in-Differences. Generalization of synthetic controls for multiple simultaneously treated units via missing data imputation: Matrix Completion.

Assignment 5.1: Estimation of temporal causal effects comparing different methods.

5.2 Generative Temporal Causal Models

Specification of temporal causal models and their unrolling. Skill estimation in the video game and sports industries. Inference flow in complete-history generative models: forward and backward propagation. Undefined messages and convergence via Loopy Belief Propagation. The forward-backward algorithm (smoothing): skill estimation over time. Temporal predictions and leave-one-out predictions: costs. Skill estimation in the history of tennis. Interpretation of odds offered by bookmakers. Simplified betting game: the honest coin. Properties of betting games. The advantage of individual diversification (negative entropy).

Assignment 5.2: Specification of a generative temporal causal betting model. Simulation and impact analysis of cooperation and individual saving.

6 Counterfactuals

Specification of models with counterfactuals, equivalence between causal inference approaches, and decomposition of total effects into natural direct and indirect effects via the mediation formula.

6.1 Models with Counterfactuals and Equivalence between Causal Inference Approaches

The levels of causal reasoning. Specification of causal models with counterfactuals: Twin Networks. Prediction of counterfactuals using factual information: Monty Hall. Revisiting the equivalence of the two causal inference approaches: Potential Outcomes and Structural Causal Models. Compliance analysis of ignorability via Twin Networks: examples. Specific constraints of Structural Causal Models: one exogenous variable per endogenous variable and determinism of endogenous variables. Breaking of Ignorability-Backdoor equivalence in general causal generative models: the paradigmatic counterexample. Generalization of the Backdoor causal inference approach to general causal generative models.

Assignment 6.1: Predicting counterfactual values using factual information.

6.2 Using Counterfactuals to Estimate Direct Causal Effects

Total vs. Direct and Indirect causal effects. Conditional Direct Effect: a contraindicated intervention. Decomposition of the total effect: Natural Direct Effect, the impact produced by treatment under an untreated mediator; Natural Indirect Effect, the impact produced by the change in the mediator under treatment. The Burks-Hammel debate: measuring gender discrimination in university admissions. Identifiability of natural effects: elimination of spurious influence between the mediator and the outcome. Mediation formula: NDE and NIE estimands from observational data and structural models. Estimation of NDE and NIE from observational data without interventions.

Assignment 6.2: Evaluation of discrimination in simulated data.

7 Decision-Making

The probability-evolution isomorphism, consequences of the multiplicative nature of variant selection, and the temporal correction of expected utility theory.

7.1 Decision-Making Over Time: The Epistemic-Evolutionary Cost Function

Von Neumann's expected utility theory. John L. Kelly's critique of Von Neumann. Betting games. The structural equivalence between betting games, hypothesis evaluation in probability, and natural selection in evolution. The advantage of individual diversification (negative entropy). The emergent properties of material forms and the advantage of variance reduction through cooperation. The fallacy of the prisoner's dilemma and the tragedy of the commons. The advantage of cooperative specialization. The Kelly criterion: cooperating with oneself. Fractional Kelly.

Assignment 7.1: Skill estimation and tennis betting.

7.2 Temporal Correction of Expected Utility Theory and Optimization as Inference

Systems of interaction with reality: action-perception. Active learning. Prediction by maximum uncertainty and action selection by minimum uncertainty. The ergodic (temporal) correction of expected utility theory. Optimal factual decision as absence of superior counterfactual alternatives. Optimization as inference. Counterfactual-based influence diagrams: the betting game and solution to specialization (individual and 2-person pool). Deterministic policies and random drip experiments. The personal lending industry case: deterministic credit allocation. Strategies for conducting micro-randomized experiments, minimizing data production and collection: the value of information.

Assignment 7.2: Micro-experiments and policy evaluation in the personal lending industry.

8 Socioeconomic Interventions

History of the colonial-modern turn, governance of the commons, analysis of current cases, and design of productive financial instruments at all levels (savings, land use planning, basic services, housing access, cycle reproduction).

8.1 General History, Commons and Successful Institutions

Limits of estimating causal effects between pairs of variables. The case of gender violence and random allocation of housing subsidies. The development of culture and agriculture. The 6 innovation centers of humanity. The colonial-modern turn: feudal pandemic, American silver, and Chinese technology. The era of genocides and the loss of cultural diversity. The deepening of the current ecological crisis and the inability of modern metropolitan science to resolve it. Characteristics of successful institutional systems in the management of the commons. Egypt: land distribution, agricultural productivity, population density and demographic dynamics. Argentina: land distribution, agricultural productivity, population density and demographic dynamics.

Assignment 8.1: Report: analysis of Argentina and Egypt data; estimates, conclusions and action proposals.

8.2 Productive Financial Instrument

Financial-productive systems in present-day Argentina: interest rate, risk and liquidity. Source of real value: water, land and life. Productive savings financial instrument — sowing pool — based on intensive agriculture and rurban development. The three actors of the system: savers, residents and administration. Savings based on pools for purchasing rural land (blocks of more than 100 ha) and interest rates aligned with rural rental values. General land planning: roads, housing, market-garden agriculture and common spaces. Specific planning of sectors and minimum size of rural areas (plots of 2000 m^2) and dwellings (250 m^2). Development of basic services: water, sewage, foundations and common area. Requirements for residents to acquire ownership: total cost and timelines for debt cancellation. The

complete cycle for process reproduction. Estimated capabilities and speed of potential growth.

Assignment 8.2: Writing the technical report and executive summary of the productive financial instrument.

Core Bibliography

- Koller, D. & Friedman, N. (2009). *Probabilistic Graphical Models.* MIT Press.
- MacKay, D. (2003). *Information Theory, Inference and Learning Algorithms.* Cambridge University Press.
- Jaynes, E.T. (2003). *Probability Theory: The Logic of Science.* Cambridge University Press.
- Pearl, J. (2009). *Causality: Models, Reasoning and Inference.* Cambridge University Press.
- Pearl, J. & Mackenzie, D. (2018). *The Book of Why.* Basic Books.
- Peters, J., Janzing, D. & Schölkopf, B. (2017). *Elements of Causal Inference.* MIT Press.
- Bishop, C. (2006). *Pattern Recognition and Machine Learning.* Springer.
- McElreath, R. (2020). *Statistical Rethinking.* CRC Press.
- Molak, A. (2023). *Causal Inference and Discovery in Python.* Packt.

Probabilistic Programming Languages

- `Pyro` (Uber AI): https://pyro.ai
- `PyMC`: https://www.pymc.io
- `Stan`: https://mc-stan.org
- `Turing.jl` (Julia): https://turinglang.org